



DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
THE NATIONAL INSTITUTE OF ENGINEERING
(An Autonomous Institution)



Project Synopsis

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Project Title: KnowledgePulse: An AI-Driven Conversational Customer Knowledge & Retention Intelligence Platform

Abstract:

KnowledgePulse ingests a company's website and uploaded documents to build a vector-indexed knowledge base and serve a RAG chatbot from it. Every conversation is logged along with a retrieval-similarity confidence score. On a scheduled cadence, an analytics engine clusters the accumulated conversations by topic and identifies recurring low-confidence questions, producing a ranked documentation-gap report the organisation can act on. The same pipeline is extended, as an explicitly-labelled proof-of-concept, to explore whether conversation content can flag customer retention risk earlier and more explainably than behavioural data alone. The system is positioned not as "another chatbot builder" but as a knowledge-intelligence layer — closing the loop between customer conversations and concrete documentation or retention action.

Objectives:

- To build a RAG-based chatbot that answers customer queries using an organisation's own website content and uploaded documents (PDF/DOCX/FAQ).
- To store every customer conversation and analyse it in scheduled batches rather than per-message.
- To automatically detect and rank documentation gaps and trending topics from recurring low-confidence or unanswered questions.
- To cluster customer queries into coherent topics using embedding-based clustering suited to short, noisy chat text.
- To generate periodic, human-readable reports that convert raw conversation data into actionable documentation-improvement suggestions.
- To explore, as a clearly-scoped proof-of-concept extension, whether conversation-derived signals can improve customer retention-risk prediction over behavioural data alone.

Software Requirements –

- **Programming Language:** Python 3.10+ (backend, NLP/ML), JavaScript (frontend)
- **Backend Framework:** FastAPI
- **Frontend Framework:** React.js
- **Database:** PostgreSQL (structured data), ChromaDB / FAISS (vector database)
- **NLP/ML Libraries:** Sentence Transformers, BERTopic, an LLM API for generation
- **Web Crawling/Parsing:** BeautifulSoup, Playwright, PDF/DOCX parsing libraries
- **Containerisation:** Docker
- **IDE:** VS Code / PyCharm
- **Operating System:** Windows 10/11 or Ubuntu 20.04+

Hardware Requirements –

- **Processor:** Intel i5 (or equivalent) or higher
- **RAM:** 8 GB minimum (16 GB recommended for embedding/model workloads)
- **Storage:** 20 GB free disk space (for vector database, models and datasets)
- **GPU:** Optional; a CUDA-enabled GPU is recommended (not mandatory) for faster embedding generation
- **Internet Connectivity:** Required for LLM API access and website crawling.

Introduction -

Organisations increasingly deploy AI chatbots built using Retrieval-Augmented Generation (RAG) to answer customer questions from their own documentation. However, the thousands of conversations exchanged with these chatbots are rarely analysed afterwards, so organisations never learn which topics their documentation fails to cover, which questions keep recurring, or which conversations were early signs of a customer's dissatisfaction. This project proposes KnowledgePulse, a Conversational Customer Knowledge and Retention Intelligence Platform in which the chatbot is only the interface — the real product is a batch analytics layer that periodically mines stored conversations to surface documentation gaps, trending topics, and, as a scoped extension, early retention-risk signals.

Literature Survey (Minimum 5 recent research Journals indexed in Scopus/Web of Science):

- Xu, Z. et al. (2024). "Retrieval-Augmented Generation with Knowledge Graphs for Customer Service Question Answering." Proc. 47th ACM SIGIR Conference. Finding: KG-augmented RAG improves retrieval and answer quality for customer-service QA over plain-text RAG. Gap: Does not aggregate recurring low-confidence topics into an org-facing report. Link: <https://doi.org/10.1145/3626772.3661370>
- Lewis, P. et al. (2020). "Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks." NeurIPS, 33, 9459-9474. Finding: Introduces the foundational RAG architecture combining a retriever with a generator to improve factual grounding. Gap: Evaluated on QA benchmarks, not conversational business settings; no mechanism to flag missing corpus content. Link: <https://arxiv.org/abs/2005.11401>
- Gao, Y. et al. (2024). "Retrieval-Augmented Generation for Large Language Models: A Survey." arXiv:2312.10997. Finding: Systematises RAG into Naive, Advanced and Modular paradigms. Gap: Notes that evaluating the usefulness of RAG outputs for business decisions, not just accuracy, remains immature. Link: <https://arxiv.org/abs/2312.10997>
- Huang, L. et al. (2023-2024). "A Survey on Hallucination in Large Language Models." ACM Transactions on Information Systems. Finding: Taxonomises hallucination causes and mitigation (including retrieval grounding). Gap: Confirms LLM-reported confidence is unreliable without an engineered proxy. Link: <https://arxiv.org/abs/2311.05232>
- Grootendorst, M. (2022). "BERTopic: Neural Topic Modeling with a Class-Based TF-IDF Procedure." arXiv:2203.05794. Finding: Combines transformer embeddings, UMAP and HDBSCAN for coherent topic clusters, outperforming LDA/NMF on short text. Gap: Follow-on studies show very short, noisy chat text still needs manual tuning. Link: <https://arxiv.org/abs/2203.05794>
- Vo, N.N.Y., Liu, S., Li, X., & Xu, G. (2021). "Leveraging Unstructured Call Log Data for Customer Churn Prediction." Knowledge-Based Systems, 212, 106586 (Elsevier). Finding: Combining structured account data with unstructured call-transcript text significantly improves churn-prediction AUC. Gap: Relies on real paired conversation-and-churn-outcome data unavailable to a student project. Link: <https://doi.org/10.1016/j.knosys.2020.106586>

Existing System & its Drawback:

- No-code RAG chatbot builders (Chatbase, SiteGPT, DocsBot, CustomGPT): store conversation logs but do not mine them for documentation-gap insight.
- Enterprise support-AI suites (Intercom Fin, Zendesk AI): offer similar "why are customers confused" insight features only within expensive enterprise tiers, out of reach for smaller organisations.
- Behavioural churn-prediction platforms: predict who is at risk using logins/billing data alone, with no access to the customer's own words, so risk scores arrive without an explainable driver.
- Generic chatbot analytics dashboards: report volume, response time and basic sentiment — descriptive, not diagnostic, and do not close the loop into a concrete fix or retention action.

Proposed System & Advantages:

- Positions the chatbot as the interface and the documentation-gap/insight layer as the actual product, differentiating from generic chatbot builders.
- Detects and ranks documentation weak spots from real recurring low-confidence questions, each backed by the underlying customer queries for human review.
- Uses retrieval-similarity as an engineered confidence proxy rather than trusting raw LLM confidence, consistent with known LLM hallucination research.
- Delivers actionable, scheduled reports instead of raw logs, making the insight usable by non-technical teams.
- Is affordable and deployable for smaller organisations that cannot justify enterprise support-AI pricing tiers.
- Extends (as a clearly labelled proof-of-concept) into retention intelligence, connecting conversational friction to an explainable churn-risk driver rather than a behavioural-only score.

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